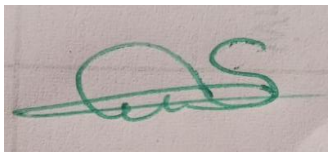
# AI-ENABLED BLOCKCHAIN ECOSYSTEM FOR SUSTAINABLE AGRICULTURAL VALUE CHAINS

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**CONCEPT NOTE SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE AWARD OF THE SCHOLARSHIP DEGREE OF MASTER OF SCIENCE IN INFORMATION TECHNOLOGY**

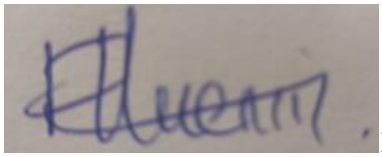
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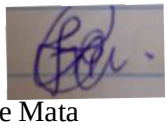
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## 1. Background and Context of Research

Agriculture remains a cornerstone of Kenya's economy, yet the sector faces structural inefficiencies, systemic opacity, and rising risk. Agriculture contributed approximately 21.8% of Kenya's GDP in 2023. However, the Kenya National Bureau of Statistics (KNBS, 2024) notes declining rigor in agricultural data collection and reporting, undermining effective decision-making.

Kenya's agriculture sector is vital for national food security. Despite increased production, post-harvest losses reach up to 40% annually (FAO, 2024). For example, maize production in 2024 reached 4,028,320 metric tons over 2,414,536 hectares, yet yields per hectare declined from 22.8 bags in 2023 to 18.5 bags in 2024 (StatsKenya, 2024). Wheat production covered only 19.6% of national demand, with imports filling the gap (StatsKenya,2024).

Fragmentation across agricultural systems farmers lacking digital identity, uncoordinated input suppliers, and siloed government data exacerbates inefficiency. Global challenges like climate change, droughts, and biodiversity loss further threaten productivity (FAO, 2024).

Emerging technologies such as AI and blockchain offer systemic solutions for integration, traceability, and transparency across agricultural value chains. Despite "smart-agriculture" initiatives, there is scarce evidence of platforms that unite all stakeholders into a unified ecosystem, connecting traceability with actionable AI insights, financial inclusion, and market access.

## 2. Research Gap

Digital agriculture initiatives in Kenya remain fragmented and pilot-based (World Bank, 2023). Existing systems target isolated value-chain challenges but fail to integrate farmers, government, financial institutions, researchers, and consumers into a single trusted network (FAO,2024).

AI-enabled decision support and blockchain-based traceability are largely underexplored in Kenya's context. Few studies examine the combined potential of AI and blockchain to improve productivity, market efficiency, and sustainability (Wambugu & Njoroge, 2023; GIZ,2023).

This gap presents an opportunity to design, prototype, and evaluate an AI–Blockchain ecosystem that enhances traceability, data reliability, and equitable participation across Kenya's agricultural value chains.

## 3. Objectives

### Main Objective

To design, develop, and evaluate an AI-enabled blockchain ecosystem that enhances traceability, data reliability, and sustainability across Kenya's agricultural value chains.

### Specific Objectives

- To design and prototype an integrated AI-Blockchain system that connects farmers, government agencies, agribusinesses, and financial institutions.
- To evaluate the system's performance in improving traceability, data integrity, and decision-making within selected value chains such as maize and dairy.
- To assess the socio-economic and environmental impacts of the system on smallholder farmers, focusing on productivity, market access, income, and sustainability.

## 4. Expected Outcomes

- A functional AI-enabled blockchain prototype that integrates key agricultural stakeholders and supports secure, transparent data exchange.
- Verified improvements in traceability, data accuracy, and decision-making efficiency within the targeted agricultural value chains.
- Quantified socio-economic and environmental impacts on smallholder farmers, including changes in income, market access, and resource utilization.
- Evidence-based recommendations to guide policy formulation and national scaling of integrated AI-Blockchain solutions in the agricultural sector.

## 5. Methodology

A mixed methods approach will be employed to integrate system design, prototype development, technical evaluation, and socio economic impact assessment. This ensures methodological alignment with the study's objectives and supports both technical rigor and real-world relevance.

### Specific Objective 1: System Design and Prototyping

- Conduct stakeholder mapping through focus group discussions and key informant engagements involving farmers, government agencies, agribusinesses, and ICT practitioners.

- Develop the system architecture that integrates AI components for predictive analytics and blockchain components for traceability and smart contract automation.
- Build a functional prototype using Python for AI model development and Hyperledger Fabric or Ethereum for blockchain implementation.

**Specific Objective 2: System Performance Evaluation**

- Deploy the prototype within selected value chains, specifically maize and dairy, to assess operational suitability.
- Collect quantitative performance metrics, including transaction time, traceability accuracy, system reliability, and data sharing frequency.
- Evaluate AI model performance using standard measures such as precision, recall, and F1-score, and assess blockchain robustness in terms of immutability and transaction integrity.

**Specific Objective 3: Socio-Economic and Environmental Impact Assessment**

- Conduct field-based surveys and semi-structured interviews with smallholder farmers participating in the pilot.
- Measure socio-economic indicators such as income changes, market access, operational efficiency, and input optimization, as well as environmental indicators including waste reduction and resource utilization.
- Analyze qualitative data using thematic analysis and quantitative data using cost–benefit and impact analysis techniques supported by SPSS or R.

**Ethical Considerations**

All research activities will adhere to established ethical standards, including informed consent, confidentiality safeguards, and compliance with the Kenya Data Protection Act (2019).

## 6. Work Plan

| Phase / Period | Activities | Expected Output Deliverable | Duration |
|---|---|---|---|
| Month 1–2 | Literature review, proposal finalization, identify gaps | Approved proposal and refined conceptual framework | 2 months |
| Month 3–4 | Stakeholder mapping and requirements analysis | Stakeholder needs assessment report | 2 months |
| Month 5–8 | System design and prototyping | Functional prototype of AI-enabled blockchain ecosystem | 4 months |

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|---|---|---|---|
| Month 9–10 | Pilot testing and data collection | Pilot deployment report with system evaluation metrics | 2 months |
| Month 11–13 | Data analysis and evaluation | System performance report | 3 months |
| Month 14–16 | Socio-economic and environmental assessment | Impact assessment report and policy brief | 3 months |
| Month 17–18 | Thesis writing and submission | Completed thesis, final defense | 2 months |

## 7. Budget Linked to Activities

| **Activity** | **Description / Purpose** | **Estimated Cost** |
|---|---|---|
| Literature Review and Proposal | Printing and minor costs | 2000 |
| Stakeholder Mapping and Data Collection | Field visits, transport, refreshments | 15000 |
| System Design and Prototyping | Software, cloud hosting, Laptop Enhancement | 15000 |
| Pilot Testing and Evaluation | Surveys, internet, minor facilitation | 5000 |
| Socio-Economic Assessment | Surveys, interviews | 15000 |
| Data Analysis and Documentation | SPSS/R, printing, binding | 6000 |
| Thesis Writing and Submission | Printing & binding | 5000 |
| Contingency | Miscellaneous | 4000 |
| Total | | 67000 |

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